

ZOO HELP*

Minimum Standards for Keeping Bears - AZA Guidelines**

Leslie A. Johnson*

INTRODUCTION

Worldwide, eight species of bears are found in a variety of habitats, ranging from the Arctic Ocean south to Borneo and the northern Andes. For purposes of this discussion, minimum standards for brown (grizzly) bear, American and Asiatic black bears, sun bear, sloth bear and spectacled bear are provided. Minimum husbandry standards for polar bears have been published by the United States Department of Agriculture. The species covered include:

COMMON NAME	LATIN NAME	WT (LB./KG)
Polar bear	<i>Ursus maritimus</i>	386 - 2000/175 - 908
Brown or Grizzly bear	<i>Ursus arctos</i>	350 - 1800/159 - 817
American black bear	<i>Ursus americanus</i>	125 - 600/57 - 272
Asiatic black bear	<i>Ursus thibetanus</i>	200 - 400/90 - 182
Sloth bear	<i>Melanursus ursinus</i>	198 - 300/90 - 136
Spectacled bear	<i>Tremarctos ornatus</i>	125 - 385/57 - 175
Sun bear	<i>Helarctos malayanus</i>	60 - 143/27 - 65

HUSBANDRY

Enclosures - All enclosures should possess a dry resting and social area, pool and den. Bears may be maintained in outdoor enclosures employing moats, thick laminated safety glass or bars. When used, dry moats should be at least 12 ft (3.7 m.) wide and 12 ft (3.7 m.) deep. Safety glass should be at least 2 in. (5 cm.) thick.

If vertical walls are used as a primary barrier, the climbing ability of bears should not be underestimated; vertical walls should be at least 12 ft (3.7 m.) high. All outdoor enclosures should have adjoining indoor facilities to permit safe cleaning and additional separation.

A dry resting and social area for one or two adult brown bears should measure at least 400 sq. ft. (37.2 sq. m.); at least 40 sq. ft. (3.7 sq. m.) should be added for each additional bear. For other species, 300 sq. ft. (9.2 sq. m.) of dry resting and social space should be provided for one or two animals, and be increased by 50 % for each additional animal. Visual barriers such as logs or boulders should be added to enclosures housing more than one animal. Regardless of species, this space shall be provided with enough shade to accommodate all individuals housed within the enclosure at the same time.

Indoor enclosures for a single brown bear should measure at least 6 ft. (1.8 m.) in width and depth, and be at least 5 ft. (1.5 m.) in height. A similar enclosure for individuals of the other species should measure at least 5 ft. (1.5 m.) in width, depth, and height. A separate holding area should be available for

each animal. Shelves or pallets should be provided for sleeping for all species except polar and brown bears. A separate indoor enclosure should be provided for each adult female housed with an adult male of breeding age. Because pregnant bears in the wild usually den in a smaller area, a recessed area reduced in size should be considered in conjunction with the primary den. Cubbing areas should be as quiet as possible.

Pool/water - Pool areas for brown bears should have a mean horizontal diameter (MHD) of at least 8 ft (2.4 m.) and a surface area of at least 96 sq. ft (8.9 sq. m.) for two adult bears. The pool should be at least 3 ft. (1 m.) deep. These dimensions do not include entry and exit areas. For each additional bear, the surface area should be increased by 40 sq. ft. (3.7 sq. m.), all of which should be at least 3 ft. (1 m.) deep.

For most other species, the pool should have a MHD of at least 6 ft. (1.8 m.) and a surface area of at least 64 sq. ft. (6 sq. m.) for two adult bears. The pool should be at least 3 ft. (1 m.) deep, both dimensions excluding entry and exit areas. For each additional animal, the surface area of the pool should be increased by 30 sq. ft (2.7 sq. m.), all of which should be at least 3 ft. deep.

Water - Fresh clean water for drinking should be available at all times. Built-in watering devices may be used for larger species; sturdy containers may be suitable for smaller species. All water containers should be cleaned and disinfected daily.

Sanitation - Hard-surface enclosures and food containers should be cleaned and disinfected daily. Resting boards and shelves where animals climb and sit should also be cleaned daily. Dirt substrates should be raked and spot-cleaned daily.

Temperature - All bears are extremely cold tolerant and can be maintained outdoors year-round. For sloth and sun bears in cold climates, hay, floor heating or infrared lamps are beneficial.

Ventilation - Indoor enclosures should be ventilated by natural or artificial means to provide a flow of fresh air.

Lighting - Because of their size, bears are usually maintained outside for a large portion of the day. When indoor facilities are used, incandescent or fluorescent lighting is a good source of broad-spectrum illumination.

Food - Bears should be fed daily. Primary diets should be based upon a nutritionally complete concentrate. Fruits and vegetables may make up to 40% of the diet by

* Omaha's Henry Doorly Zoo, 3701 South 10th St., Omaha, NE 68107, USA

weight, depending on whether dry or canned food is the base diet. Brown, black and polar bears may receive fish, fed at approximately 20% of the overall diet. Frozen fish should be supplemented with thiamin (25 mg/kg food) and Vitamin E (100 IU/kg food) immediately before feeding.

Bears should consume 1.5 - 2.0 % of their body weight per day in dry matter but there may be significant seasonal fluctuation in food intake. In addition, it may be difficult to assess quantities since many foods (fruits and vegetables) are high in water content. Bears should consume food proportionally although in practice they usually prefer high moisture foods and may leave some dry food uneaten. If this occurs, the amount of high moisture food should be reduced to encourage consumption of commercial foods. Dry foods may be softened with water to encourage or facilitate consumption, particularly if dental problems are present.

Fasting - During winter, bears housed in northern climates, as well as pregnant females in all regions, may hibernate or become inactive for periods of 2 -3 months. Food intake will decline dramatically or stop altogether and individuals can cease being fed during that period. As the bears emerge from hibernation, food should be gradually increased as appetites improve.

Social - Bears are generally considered to be solitary animals, coming together only for breeding or in situations of extreme food abundance. The only exception is the female/offspring social unit. Since bears are solitary most of the time, they can be kept in units of one. However, bears may also be housed in pairs, or in larger groups containing a single male. Expectant females should be moved to isolated cubbing dens prior to birth.

Veterinary care - Services of a veterinarian experienced with large carnivores is recommended. Because bears are more prone to internal parasites than some other large mammals, fecal exams should be performed twice a year and treated appropriately. No general vaccinations are recommended.

On an opportunistic basis, bears should receive physical exams that include ophthalmic examination, dental cleaning and calculus removal, claw examination and trimming, abdominal palpitation and blood collection for a complete blood count and serum chemistry panel. Individual bears involved in management programs or otherwise likely to be transferred should be permanently identified with tattoos of their studbook or accession number. Insertion of a Trovan microchip transponder at the base of the left ear is also recommended.

ENRICHMENT

Techniques to enrich their environment should be explored. Because eating is the most stimulating part of a bear's day, modification of feeding routines by changing the time of feeding as well as by hiding small food items within the enclosure will encourage natural foraging behaviors. In addition to the main feed, supplemental feeding solely to increase activity can be added to the daily routine. These feeds (vegetables, fruits, browse, rawhide dog bones, and live or fresh fish) may supplement the main feeding and be delivered at any time to keep the bear active. Food may be finely chopped and scattered/hidden throughout the

enclosure, frozen in a block of ice or enclosed in containers that can be manipulated.

Bears may also be provided with objects to manipulate or explore. Such objects should be sturdy and non-toxic; examples include traffic cones, "boomer" balls, heavy rubber buckets, etc. To maintain novelty, they should be rotated daily or periodically. Because bears enjoy digging, patches of bare soil or grain husks may be added for novelty as well as an additional area for hiding foods.

ADDITIONAL LITERATURE

Chapman, Joseph A. and George A. Feldhamer 1982. WILD MAMMALS OF NORTH AMERICA: BIOLOGY, MANAGEMENT, AND ECONOMICS. The Johns Hopkins University Press, Baltimore. Pp.504-566.

Domico, Terry 1988. BEARS OF THE WORLD. Facts on File Press, NY, NY. 189 pp.

Hall, E. Raymond 1981. THE MAMMALS OF NORTH AMERICA, VOL. II, SECOND EDITION. John Wiley & Sons, NY, NY. Pp.947-960

Macdonald, David 1985. THE ENCYCLOPEDIA OF MAMMALS. Facts on File Press, NY, NY. Pp.86-97.

Nowak, Ronald M. 1991. WALKER'S MAMMALS OF THE WORLD, 5TH EDITION. Johns Hopkins University Press, Baltimore, Pp.1083-1095

Stirling, Ian 1993. BEARS: MAJESTIC CREATURES OF THE WILD. Rodale Press. Emmaus, PA. 240 pp.

* Introducing a new column, ZOO HELP. This column will consist of short articles and notes which have been assembled for a larger zoo audience. Some of these pieces will be from zoo associations in other parts of the world and may contain inappropriate, impossible or unworkable suggestions with regard to procurement of food, climatic factors, economics, etc. We have two batches of material ready from the AZA Minimum Standards and from the AZA Resource Centre. We will use material from other associations also. This column is to insure that something useful to zoological parks and gardens or captive breeding centres appears in ZOOS' PRINT every issue.

** AZA stands for the American Association of Zoos, which has the largest association of zoo professionals in the world and perhaps the most active association. The AZA Minimum Standards were prepared at the request of the US Department of Agriculture, which is the nearest equivalent to the Ministry of Environment and Central Zoo Authority in USA, as a way for their inspectors to better evaluate zoos, dealers, road-side facilities and anyone else requiring a legal permit. They asked for primate standards in the mid-1980's, and all others about 1990. I oversaw the project from the early 1990's until completion in 1997 when the printed version was distributed to all AZA zoos. The standards are minimal in many respects, but optimal in others. While it may be safe to say they were developed for temperate zoos, contributing zoos such as San Diego, Brownsville and Miami not temperate so there may be little difference climatically.

